Response to referee comments: PRD Submission DM13138,

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Author response to the referee report for the paper DM13138, entitled *'Adaptive cancellation of mains power interference in continuous gravitational wave searches with a hidden Markov model'*.

We thank the referee for their constructive review. We agree with all the comments and have made the recommended changes as described below. The referee comments are reproduced in italics for convenience. We separate the response into "main" comments (indexed alphabetically) and "additional" comments (indexed numerically), following the convention of the referee.

Referee comments

Main comments

A. *In section II, it is not sufficiently clear if the model is intended as a realistic representation of the North American power grid, or just a phenomenological model that is "good enough" for the purpose of subtraction. This should be spelled out better. An authoritative reference on the actual grid's design and/or statistical properties, if the authors can find one, would also be highly valuable for the typical target audience of this article (astrophysicists / data analysts) who are unlikely to have such information at hand.*

Author response:

Yes, we agree with the referee that it is important to specify clearly that the model is intended as a phenomenological model that seeks to reproduce the main qualitative features of the mains power (i.e. constant central frequency with small amplitude stochastic wandering, see third paragraph in Section IIB.). We have followed the referee's suggestion and updated the discussion in Section II to make this point explicitly.

TK: I cannot find any references for the empirical behaviour of the grid. Perhaps the EE-side might know.

B. *In the end, it is not clear enough how big the positive impact of the method really is. Especially, and regardless of the answer to point A, it seems necessary to present at least some level of real-data test of the proposed method, beyond the presented simulated noise case. Since LIGO data is readily available via the GWOSC site, this should be straightforward to perform. A clearer summary of the results in abstract and conclusion would then be appreciated.*

Author response:

Yes we agree that a clearer summary of the main results quantifying the impact of the method is important to include. We have updated the abstract and the conclusion accordingly. Specifically

- For the representative system (Table 1) ANC suppresses the 60-Hz interference by approximately 20 dB.
- Without ANC, the HMM tracker is unable to track the wandering CW signal. After ANC, the HMM tracker is able to track the wandering CW signal. The time-averaged RMS error in the HMM tracking after ANC is 0.038Hz for low σ_f and $0.47 Hz$ for high σ_f.
- The detection probability at a 5% false alarm rate, $p_{\rm d}(0.05) \approx 0.5$–$0.6$ across a range of mains power interference parameters and filter parameters.

TK: Tests on real-data to be discussed in person. A summary of some thoughts below.

We currently generate some synthetic strain-channel data $x(t) = h(t) + c(t) + n(t)$, and some synthetic reference-channel data $r(t)$, according to some analytical expressions which are specified in the manuscript.

Which of these components would be appropriate to replace with real GWOSC data?

We don't have GWOSC $h(t)$ - no CWs have been detected! Let's assume that in some strain data $x(t)$ that we obtain from GWOSC that there is no $h(t)$ signal in there. The procedure for testing the method on real data might then look like:

- Take some strain-channel data $x(t)$ from GWOSC
- Simulate some synthetic $h(t)$ according to the manuscript equations.
- Inject the $h(t)$ into $x(t)$ to obtain some data $y(t) = h(t) + x(t)$ which gets passed to the ANC scheme.
- Take $N \leq 9$ PEM channels from GWOSC. Pass these to the ANC scheme as $r_{\rm empirical}(t)$.
- Run ANC scheme - can we recover our injected $h(t)$?

Additional comments

1. *The abstract currently points out that the validation test is on "synthetic signals", which is perfectly fine, but what is more important is the nature of the noise data.*

   TK: this is related to point B, re real data. To be updated once point B has been answered

2. *A brief summary of the results would be valuable at the end of the abstract (as per point B above).*

   Author response:
   Yes, we agree that the main results should be better summarised in the abstract. We have updated the abstract accordingly.

3. *A few comments on references the introduction section: The first 3 are very old, describing glitches in initial detectors - consider updating to 2G references like the Davis+ papers already cited later. The standard Advanced Virgo detector reference arxiv:1408.3978 seems to be missing. Please check if [21] (Lee et al) was indeed the intended citation - this seems to be a deep learning waveform model paper, not about artifact identification. At the end of the second paragraph, another important class of line vetoes not cited yet are those based on multi-detector consistency (e.g. doi:10.1088/0031-8949/90/12/125001 and doi:10.1103/PhysRevD.89.064023)*

   Author response:
   We thank the referee for bringing to our attention references which we had previously overlooked, and the erroneous [21] Lee et al. citation. We have now included all the suggested references, removed reference [21] , and included discussion on the class of vetos based on multi-detector consistency

4. *In the paragraph below, the statement that candidates are vetoed blindly without checking the strength of the noise contribution is not true for all standard CW pipelines. More nuanced discussions can be found in some recent LVK results papers.*

   Author response:
   Yes we agree with the referee that the previous statement was too broad. We have now updated the paragraph to caveat that some pipelines do have "line robust" algorithms which combine statistics from different detectors to suppress single detector artifacts (e.g. Keitel et al. 2015, Keitel 2016a,b).

5. *Please try using multiple linestyles for better readability of figures 1,6,8,9. In particular, in Fig. 6 it is difficult to discern that the blue $x(f)$ curve actually shows the signal peak, as it is mostly hidden behind the green $h(f)$.*

   Author response:
   We have updated all linestyles in the figures to improve readability, as suggested.

6. *At the end of II.A, there should be an astrophysical reference for the Crab's frequency, not just a software paper. (Or both together, since citing software used is indeed commendable.) Since the Crab's frequency evolves relatively quickly, a reference year would also be appropriate when quoting it with decimal digits.*

   Author response:
   We have updated the reference to the Crab frequency to include both an astrophysical reference and a software reference. We now also quote the Crab frequency for a given reference year, as suggested.

7. *Fig. 1: missing definition of the abbreviation "ASD"*

   Author response:
   The abbreviation ASD is now defined in the caption of Figure 1.

8. *After Eq. (1) the strict separation of frequency and amplitude modulation is misleading, as both parts of the Earth's motion contribute to both modulations.*

Author response:
The text has been updated to remove the misleading separation between the two modulations.

9. *Before Eq. (4), when talking about the coupling "in various complicated ways", would this not naturally be expected to cause stochastic overlaps of components with different time delays tau? Could the authors comment on whether this is a likely explanation for the observed losses in coherence?*

10. *footnote 3: gitlab file URLs can be quite ephemeral. Is there no better (more stable) way to reference this list, such as a GWOSC release with DOI, a LIGO DCC entry, etc?*

11. *The statements at the end of II.C and in the Fig. 4 caption contradict each other: are the higher-frequency peaks harmonics of the power mains, or "of different provenance"?*

12. *In the coherence discussion, it remains unclear what threshold is considered to be still sufficient, and below which the channel would no longer be considered a reliable witness.*

13. *In Eq. (12), it is not made explicit what argument the "arg min" is taken over. Would it be appropriate to write $e_k(w)$ with an explicit argument?*

14. *In IV.A, can the authors make an argument why neglecting the Earth's movement is not changing qualitatively the signal-vs-line behaviour? Since CW signals are expected to modulate very slowly intrinsically, typically the shape and extent of their detector-frame frequency spectrum should be dominated by these terms.*

15. *Also in the same discussion, there is a reference to the F-statistic. This is taking a maximum likelihood over amplitude parameters, not over those related to the Earth's movement. Indeed F-statistic searches typically also include a "demodulation", which is probably what the authors had in mind here. But the way it is currently phrased will confuse a non-expert reader into thinking it maximizes over frequency evolution parameters.*

16. *In IV.B, the definition of signal-to-noise ratio does not match the standard in the CW literature, which usually includes the observation time. The duration is also important for the frequency resolution. Is it the 800s visible in Fig. 7? Please clarify more explicitly.*

17. *The case (ii) "not plotted for brevity" would seem relevant to show - since now fgw lies within the affected frequency range, it sounds like it could be qualitatively different, and more impressive to demonstrate that the method can still deal with it?*

18. *At the end of IV.B, have the authors verified that the Earth Doppler effect is negligible compared to these frequency errors?*

19. *In figures 8/9, it would be interesting to also compare with the performance without ANC - is that similar to, or even worse than the "random guess"?*

20. *Why is the "zero interference ROC curve" shown only in Fig. 9, not 8? Is it understood why the best performing ANC results are still quite far away from this, if before the subtraction looked visually very impressive?*

21. *Near the end of V.B, latency issues are discussed. Are these really relevant for CW searches, which are usually run on archival data?*